

Decoda Engineering Skills Test: LiDAR Processing

Overview

LiDAR (“light detection and ranging”) sensors map the distance to nearby objects using pulsed laser beams, allowing autonomous vehicles to perceive their surrounding environment in real-time.

Your goal: Create a ROS 2 node that processes raw LiDAR data to differentiate between **ground** and **obstacles**.

Challenge

Implement a point cloud segmentation node with the following specification:

1. **Subscribe** to the `/lidar` topic (`sensor_msgs/PointCloud2`).
2. **Segment** the point cloud into ground and non-ground (obstacle) points.
3. **Publish** the results to two topics:
 - `/ground` - drivable surface
 - `/nonground` - potential obstacles
4. **Visualize** the results, displaying ground and obstacles in distinct colours (e.g., brown for ground, purple for obstacles). RViz2, Foxglove, or any other visualiser is fine.

Note: A simple segmentation is sufficient - further clustering or filtering is not required.

Extension: Reliability

Optional, but the strongest submissions will address it. Our nodes run unattended on machines in a pit, where nobody is watching a terminal and nobody can reach the hardware quickly.

Give us reason to trust yours:

- Demonstrate that it behaves correctly, and that your evidence would catch it if it stopped behaving correctly.
- Make its behaviour and health visible while it is running, so a failure is noticed rather than discovered later.

Keep it self-contained: nothing hosted or external, only what runs on your own machine. Implement as much as you can; where you don't, discuss what you would do instead and why. A well-argued discussion is worth more than a token implementation.

Constraints

Constraint	Detail
OS	Ubuntu 24.04 (native, WSL, or Docker)
Middleware	ROS 2 Jazzy
Language	C++ or Python
Libraries	Open-source libraries are permitted, provided you can explain their operation

Performance: Your node should be designed to support real-time processing. While the test environment may not reflect this, your design choices should demonstrate the capability.

AI/LLM tools: Usage of AI tools is permitted. If used, please explain how they were utilised and list the specific prompts used to generate code or solutions.

Test Data

A rosbag2 recording is provided for testing your solution.

File	Description	Download
<code>lidar.tar</code>	A short rosbag2 (MCAP) recording of a haul road.	Attached in email

Deliverables

Submit a single zip containing three things.

1. Source code

Clean, structured code for your node, with its git history intact. Include the `.git` directory, and please don't squash the work into a single commit.

2. A report

A short written document covering:

- **Build instructions** - how to build and run your node.
- **Design overview** - the design choices you made and the trade-offs you considered.
- **Assumptions** - anything you assumed during development.
- **Dependencies** - any open-source libraries you used.
- **AI/LLM usage** - if applicable, how you used AI tools and the specific prompts used.
- **Improvements** - what you would do given more time.
- **Reliability extension** - if attempted, what you implemented, plus the discussion described above.

3. Recordings

Your visualiser showing the segmentation, along with anything else you want to demonstrate running.

Submission

Please respond to the email from which you received this skills test, attaching or linking to your zip file.

Note: If you are unable to complete the task, please submit a written report describing what you accomplished and the specific challenges that prevented further progress.